Siqi Chen

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Seeking Research Assistant (RA) or research internship opportunities in 2027.

Research interests: Dexterous manipulation, robot learning, vision-language-action models, and reinforcement learning for robot control.

Education

Harbin Institute of Technology (HIT) — Harbin, China
M.Eng. in Control Theory and Control Engineering (School of Astronautics) — *Sep 2025 – Present*

Harbin Institute of Technology (HIT) — Harbin, China
B.Eng. (School of Astronautics) — *Sep 2021 – Jun 2025*

- GPA: 91.21/100 (overall rank: top 15%).
- Coursework: Calculus (98), Linear Algebra (96), Probability & Statistics (94), Automatic Control Theory (95.8).
- English: CET-6 507.

Research Manuscript

ResMerge — Under Review
Fourth author

- Conducted small-model ablation experiments on LIBERO-Object under a single RTX 4090 setup; implemented evaluation scripts for result aggregation and plotting.
- Created core paper figures (teaser and method diagram) to improve clarity of the submission.

Experience

PsiBot
Robotics Engineering Intern — *May 2026 – Aug 2026*

- Developed exoskeleton-glove teleoperation and motion retargeting for multi-DoF dexterous hands, including Wuji (20 DoF) and DH01 (21 DoF).
- Integrated ROS2 glove inputs, kinematic retargeting with joint constraints, RViz visualization, and trajectory smoothing for hardware commands.
- Prototyped an ego-view video-to-robot pipeline for hand trajectory reconstruction and motion retargeting; end-to-end integration remained incomplete at the end of the internship.

PKU–PsiBot Joint Lab — Beijing, China
Research Intern — *Nov 2025 – Feb 2026*

- Delivered project-oriented engineering support for embodied robotics research, including system integration and experiment tooling.
- Implemented and maintained training/evaluation workflows; improved reproducibility and iteration speed for benchmark experiments.

Standard Robots — Shenzhen, China
Embodied AI Intern — *Jul 2025 – Sep 2025*

- Built robot teleoperation and data collection workflows; produced structured logs for downstream learning or debugging.
- Participated in debugging DARWIN-01 electrical bus issues and contributed to NVIDIA Jetson-based development; gained hands-on understanding of the full robot system stack.

Selected Projects

Exoskeleton-Glove Teleoperation for Dexterous Hands | *ROS2, Retargeting* 2026

- Mapped exoskeleton-glove motion to Wuji (20 DoF) and DH01 (21 DoF) hands, bridging human and robot hand kinematics.
- Adapted optimization-based retargeting using Pinocchio forward kinematics and NLopt SLSQP, with fingertip and pinch objectives, joint constraints, and regularization.
- Connected ROS2 glove inputs to RViz and hardware command interfaces; integrated Ruckig trajectory smoothing and joint-limit handling for robot-hand control.

Ego-View Video-to-Robot Pipeline (Prototype) | *Hand Trajectories, MuJoCo* 2026

- Prototyped world-frame bimanual hand trajectory reconstruction from egocentric RGB video, combining hand pose estimation, camera tracking, and metric depth.
- Worked on robot-motion retargeting and MuJoCo replay; end-to-end integration remained incomplete at internship handoff.

GR00T-N1.6 Fine-tuning on LIBERO | *DeepSpeed ZeRO-2, BF16, LIBERO, VLA* 2026

- Fine-tuned NVIDIA GR00T-N1.6 (3B VLA) on LIBERO (4 suites, 40 tasks) using an end-to-end training and evaluation pipeline.
- Applied a parameter-efficient recipe: froze vision encoder and LLM; trained the projector and diffusion decoder ($\sim$40% trainable parameters).
- Achieved 97.8% average success (782/800 episodes) and released 4 benchmark-validated checkpoints on HuggingFace.

RDT Fine-tuning and RLinf Integration | *LIBERO, Diffusion Policies* 2025–2026

- Built supervised fine-tuning and evaluation workflows on LIBERO; cached T5-XXL instruction embeddings to improve training throughput by $\sim$30%.
- Compared full fine-tuning (76–94% success, suite-dependent) with LoRA ($\sim$20%); profiled automated checkpoint evaluation.
- Integrated RDT checkpoint evaluation into RLinf with 9-DoF joint proprioception. PPO-style fine-tuning remained incomplete because diffusion action log-probabilities were not yet supported.

Technical Skills

Learning & Embodied AI: Reinforcement Learning, Imitation Learning, Diffusion Policies, VLA fine-tuning, Benchmarking & Ablations
Robotics: ROS2, dexterous-hand teleoperation, motion retargeting, trajectory smoothing, data logging, LIBERO, RViz/MoveIt2
ML/Systems: PyTorch, DeepSpeed, CUDA, distributed training/evaluation, wandb
Programming: Python, C++, Linux, Git

Awards

Silver Medal, 2025 World Humanoid Robot Games (Material Sorting Track)
Operator & Algorithm Developer *Aug 2025*